QUT

Project Plan Template

Introduction to Project Management

Project title	Logickube Customer Journey Case Study
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1. Project Background

Logickube is a company that aims to support eCommerce companies by making informed data-driven decisions to increase business success and profit. QUT has partnered with them for the capstone unit of MXB348. Logickube have provided us with two synthetic datasets containing the journey and conversion rate of a customer, along with objectives that provide us with opportunities to hone our data modelling skills on tasks that reflect real-world scenarios.

2. Service Need / Strategic Need

Our benefit is that we get to engage with a professional company and gain skills in project management. The lecturers and tutors do not provide any “handholding” either, so it requires us as students to problem solve and apply our skills in statistics and data science to achieve the necessary tasks.

A long-term benefit would be that the final report may be a useful document to show off when applying for job/research positions within the fields of statistics and data science.

For Logickube, their benefit is potentially innovative models or methodology that help them improve their techniques. This may lead to greater profit for Logickube as more companies invest in them.

3. Objectives

Task 1: Anticipate future customer conversion behaviour through predictive modelling. Two predictive models are required.

- The probability of a customer being converted on a return visit.
- At any given stage of a customer journey, predict how many steps it will take to reach a conversion.

Task 2: Based on the findings from Task 1, provide customer life cycle labels and justify.

Task 3: Use an attribution method to determine the 'credit' of a contributing factor towards business success, considered as a conversion. The attribution results should be able to justify the contribution of the following grouped factors:

- Each individual channel (i.e. Organic, Google Search, Facebook)
- Journey Length (i.e. number of touch points, time a journey takes, collapsed touch points)
- Channel groups (i.e. Google paid = Google Search + Google Display + YouTube)

SMART:

- Specific: A model that forecasts probability for customer conversion depending on attribution pathways via channel touchpoints.
- Measurable: Quantitatively conveys information relative to the goals.
- Achievable: Yes. Project scope is of manageable size whilst covering a broad range of topics.
- Relevant: Predictive model and methods should be applicable and easily extendable to similar datasets about customer conversion advertisement pathways.
- Time-bound: Requires completion within 3-month period.

4. Assumptions, Exclusions & Constraints

Some of the assumptions relate to the interpretation of Logickube's language used in the task description.

Other assumptions relate to the limitations of the data.

- Customers can only interact with the channels provided in the dataset.
- Each conversion has equal weight.
- Customers are isolated from one-another.
- The end of one of session and the start of another for the same customer is defined by leaving/turning off their device.
- The time between sessions has no effect on the journeys or customer.
- All assumptions held by our model of choice will be upheld through our use of said model (Markov chain memoryless property, etc.)

Exclusions:

- Long-term outcomes will be unknown, so there won't be any knowledge on if product is a success.
- Detailed interactions between repeated journeys from the same customer will not be investigated.
- No further investigations into what type of interaction both "Organic" and "Direct" contacts are. If the clients were seeking more value from the data perhaps understanding how these states are categorised would be useful.

Main constraints of the project:

- Only 3 people in the group who cannot work full-time on the project.
- The project is required to be completed within a 10-week period.

5. Approach

The approach to the project will start linear and expand to become more iterative, especially after the predictive models in Task 1 are complete. Once Task 1 is complete, each member will choose sections of the remaining tasks and report sections.

The goal would be to build prototypes and continue to improve on them until the final model is satisfactory to us and the marking criteria.

High-level Project Lifecycle – consistent communication and contribution between all members, which include new ideas and perspectives. Proof-reading and model testing on each other's prototypes to ensure reproducible models and methods.

6. Scope — Product Breakdown Structure (PBS)

A *Modelling Report* is the main deliverable which contains a literature review and methodology involved for the results we have produced to the tasks listed in Section 3. The report also requires an executive summary that is a streamlined and less technical iteration of the modelling report.

A P*reliminary Modelling Report* and *Presentation* based on the report work are also required prior to submission of the final. These will be useful in receiving valuable feedback from the industry members and university lecturers and tutors to ensure the *Modelling Report* is high quality and thorough.

All members will be involved.

7. Scope — Work Breakdown Structure (WBS)

Start – group assigned.

Download data and initialise GitHub repository.

Begin logging group session progress and aims.

Task 1 – predictive modelling.

Clean data and perform feature engineering.

Conduct literature review into useful predictive modelling techniques.

Begin testing predictive models by following Pareto Principle.

- Task 1 will be worked on by all three group members as it is expected to be the task with the heaviest workload.

Continue testing until results of sampled group are justifiable.

Task 2 – customer life-cycles.

Conduct literature review on life-cycle labelling based on techniques used predictive modelling.

Ensure that life-cycle labels stay within the constraints of the assumptions.

Verify labels are applicable to all customers.

Task 3 – business success attribution.

Conduct literature review on attribution models.

Develop attribution model until it justifies the contribution of each factor grouping.

Final products – presentation and report.

Compile all methods and findings into final report.

Present and explain project in front of stakeholders.

8. Organisational Breakdown Structure (OBS)

The members of the team are:

- Colby
- Retief
- Sebastian

The support members for the project are:

Shen Liu – Logickube CEO who is the representative to reach out to for the company.

Travis Keir – Logickube employee and former QUT student who completed the same unit.

James McGree – unit coordinator, helpful support for the mathematical side of project.

Alex Liu – unit tutor, also helpful support for the mathematical side.

Robert Hoffman – Project management guidance.

9. Responsibility Allocation Matrix

Task	Project Team	Industry Support	University Support	Project Management Support
Team Logs	R	-	I	C
Project Management Plan	R	C	I	C
Preliminary Report	R + A	C + I	C + I	-
Final Presentation	R + A	C + I	C + I	-
Final Report	R + A	C + I	C + I	-

R = Responsible | A = Accountable | C = Consult | I = Inform

10. Key Milestones and Schedule

The critical activity of the first half of the entire project is cleaning the data and creating an initial model. In essence, the first two weeks were roughly dedicated to understanding the data and listening to the clients and understanding the value they wanted to derive from that information. Initially, there was a lot of focus on individual research, this was just to flesh out the project and understand how the different minds of the team members approached the posed problem.

The following two weeks were than primarily aimed at converging on big picture approach and breaking down the steps to reach a valid conclusion. It was at this point that we collectively agreed that a Markov Chain approach to the client's problem would be the most valuable.

In the most two recent weeks, we completed an initial 200-line algorithm which smoothly runs an initial model on the provided data. This resulted in us recently having a transition probability matrix with our current dataset and from here we plan on using the matrix as a primary means to reach a conclusion about customer conversion

Over the next week, we will work on a preliminary report, that describes our process and current model. This will then extend to model refinement and extensions to fulfil the provided objectives. Once the model is complete (or close to completion), we will shift our focus to report writing. For week twelve, we will transform our report and outcomes into a succinct presentation for the stakeholders. Week thirteen will then completely consist of cleaning up the report.

11. Budget

The biggest estimated cost for the project will come down to prioritising and reallocating our time from current university study loads to this project. Mostly, this means that the costs for this project were primarily the opportunity cost of not being able to use the same amount of time to work on projects and assignments for other courses.

12. Risks and Issues Management

Small Group Size → as a result of having a small group size of three, losing a team member could result in a workload that is too heavy for the remaining members, leading to inability to complete scope of tasks within the expected time frame.

Data Loss → if updates on data cleaning and modelling procedures are not properly backed up (i.e. files uploaded to the GitHub repository), then it increases the time taken to complete projects.

Individual In attendance → the scheduled time to meet group members was not always optimal, unpredictable circumstances meant that there were occasional sessions where individuals were not able to attend meetings

Software Limitations → library accessibility, different hardware and software's will interact differently with the presented data. This means that although one person could run the full line of code on their PC this doesn't necessarily mean that it would work smoothly for someone else.

Conflicting Propositions → the group must come to a complete agreement before each decision is made. Conflicting ideas take time to resolve.

Confidentiality Limitations → the data and outcomes have some level of confidentiality associated with them. External assistance is not always possible.

Constrained Timeline → having a constrained timeline limits the depth and scope of our responses. There may be better conclusions or further model development that can't be completed before the deadline.

13. Stakeholder Management

The first major stakeholder is Logickube. They provided the project scope that they expect can be completed within the time frame. A major expectation is that we as a team handle the data provided to us with professional care – public sharing is prohibited.

Another major stakeholder is QUT. The quality of the unit depends on our engagement and performance with the project. It is expected that the highest accreditation is achievable by following the marking criteria. It is because of this that they are the stakeholder of *greatest interest* to us.

The group members are stakeholders. This project is an opportunity for each of us to demonstrate our individual capabilities to other stakeholders and develop our skillsets.

14. Quality Management

Verification: The quality of models will be determined by how well the sampling group reflects the results of the tested model. Consultation with the unit lecturing team will also be essential to ensure that we have followed a sound mathematical procedure.

Validation: the preliminary report and final presentation will both provide us with feedback so for assurance that we are on the right path.

- Reproducibility of model methodology and results would be something that I would expect to be confirmed from the unit lecturing team.
- It is assumed that the model's purposefulness would be measured by Logickube and they would provide feedback on how our procedure compares to those they have used. This is because they would have all the information on the actual impact of methods on business success of real companies.

15. Implementation Plan

Once this project plan has been submitted, the project team will turn their focus towards development of a preliminary version of the report.

Once the project plan is implemented, we will look towards creating a touch point proportionality algorithm that will provide detailed information about the relationship between customer step journey and the proportion of customers that are converted at that equivalent stage. This will be the first milestone and from here we will begin to work on the final report and extract relevant findings and conclusions.

16. Lessons Learned

Within this report we have outlined our approach and current situation. Within this report are the risks and issues that we have and will expect to face in the future. In essence, this project planning has been a reflective task that has allowed us to extrapolate and simulate possibilities and evaluate outcomes based on our current point in the project. This process of reflection has been a valuable exercise that we shall continue to exercise, because it segregates necessary accountability and responsibility of work.